Daily Progress Report

Results and Physical Interpretation of Air–Sea Heat Flux Drivers (1993–2023)

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1 Objective

This report summarizes the completed exploratory data analysis (EDA) phase for turbulent air–sea heat flux modelling over the region 40°E–120°E and 60°S–30°N using ERA5 reanalysis data.

The objective is to:

- Quantify thermodynamic gradient formation.
- Validate bulk aerodynamic flux formulations.
- Identify dominant drivers of observed fluxes.
- Diagnose missing physics via residual analysis.
- Establish scientific motivation for a Physics-Informed Neural Network (PINN).

2 Driver Influence on Thermodynamic Gradients

2.1 Driver vs Δq and ΔT Correlation

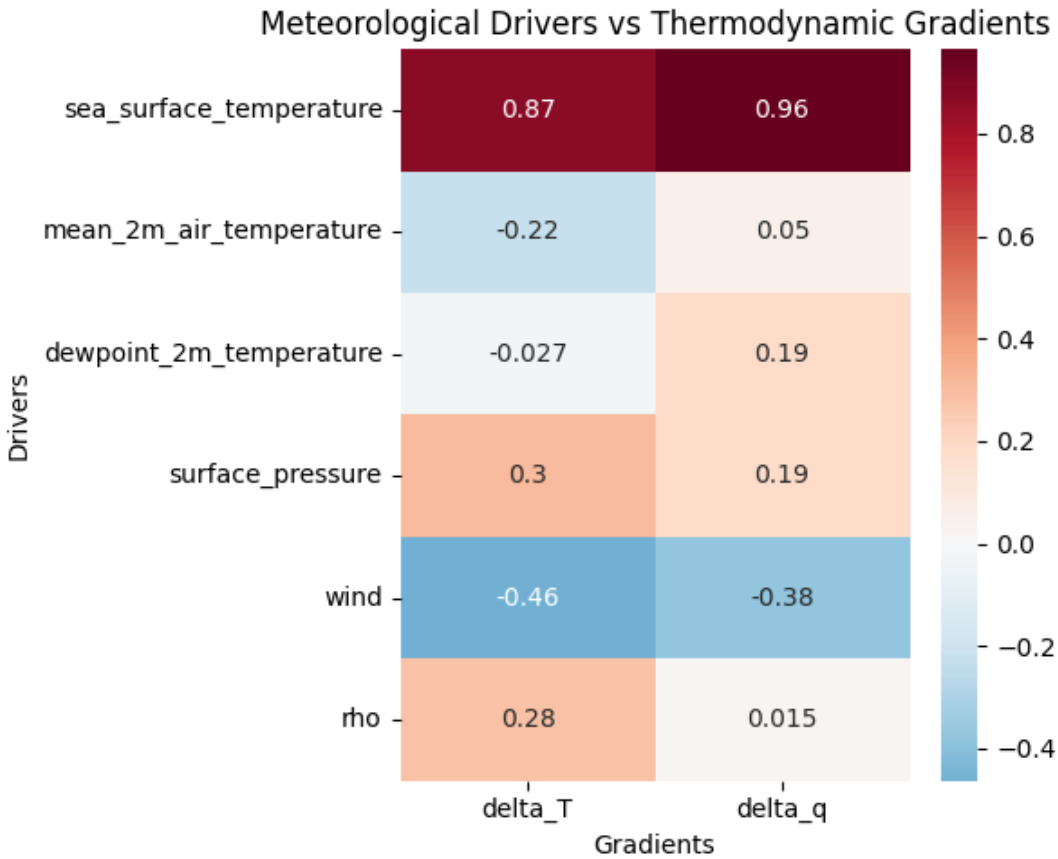

Figure 1: Correlation between meteorological drivers and thermodynamic gradients (Δq, ΔT).

Sea surface temperature (SST) dominates gradient formation, confirming the ocean as the primary thermodynamic reservoir.

Wind speed exhibits negative correlation with gradients, reflecting mixing-induced gradient erosion.

3 Validation of Bulk Aerodynamic Physics

3.1 Driver vs Bulk Flux Correlation

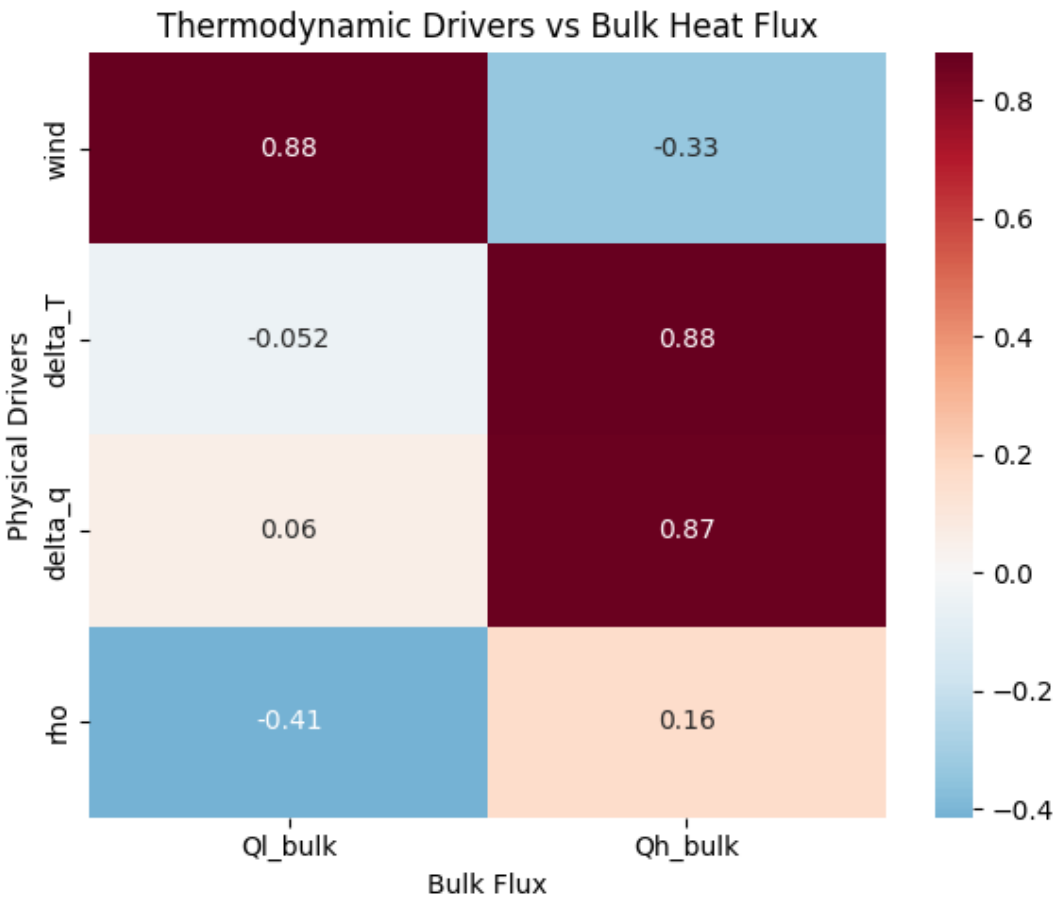

Figure 2: Correlation between drivers and bulk-estimated heat fluxes.

Bulk latent heat strongly correlates with wind and humidity gradient, while bulk sensible heat is primarily controlled by ΔT.

This confirms correct physical implementation of bulk formulations.

4 Drivers of Observed ERA5 Fluxes

4.1 Driver vs Observed Flux Correlation

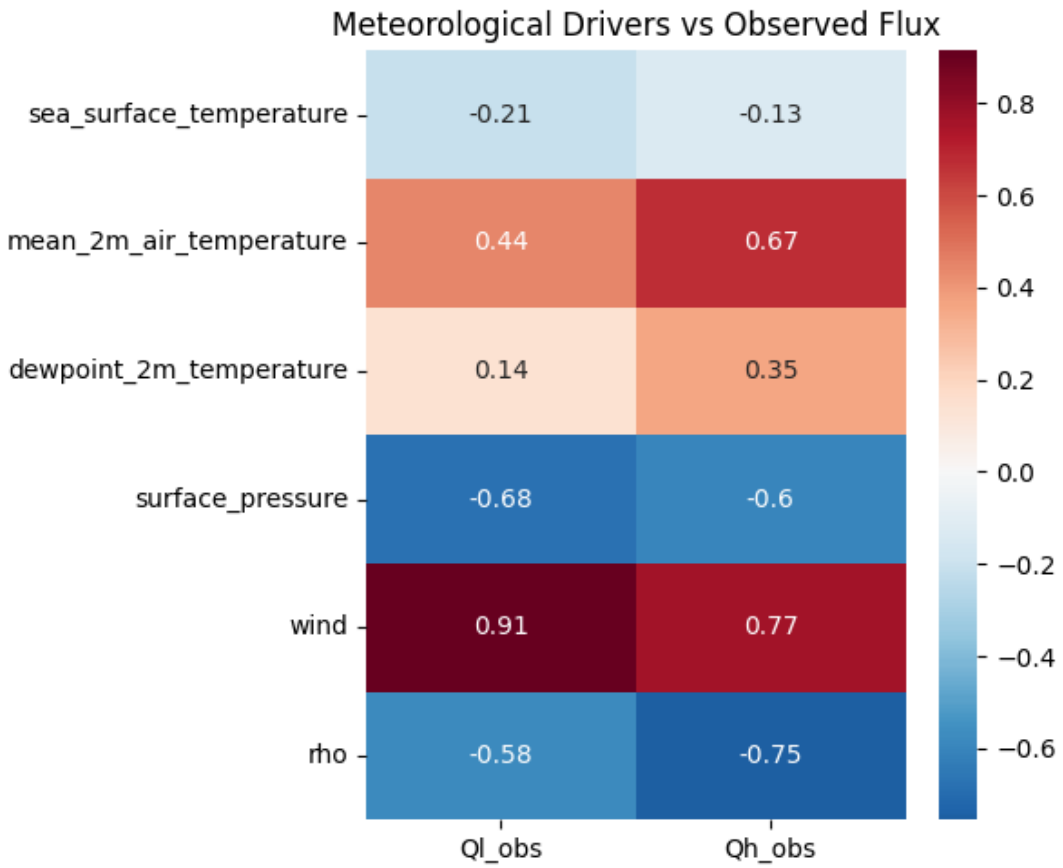

Figure 3: Correlation between meteorological drivers and observed ERA5 turbulent heat fluxes.

Observed fluxes show additional sensitivity to atmospheric density and surface pressure, indicating boundary-layer influences not fully captured by bulk formulas.

5 Bulk vs Observed Flux Skill

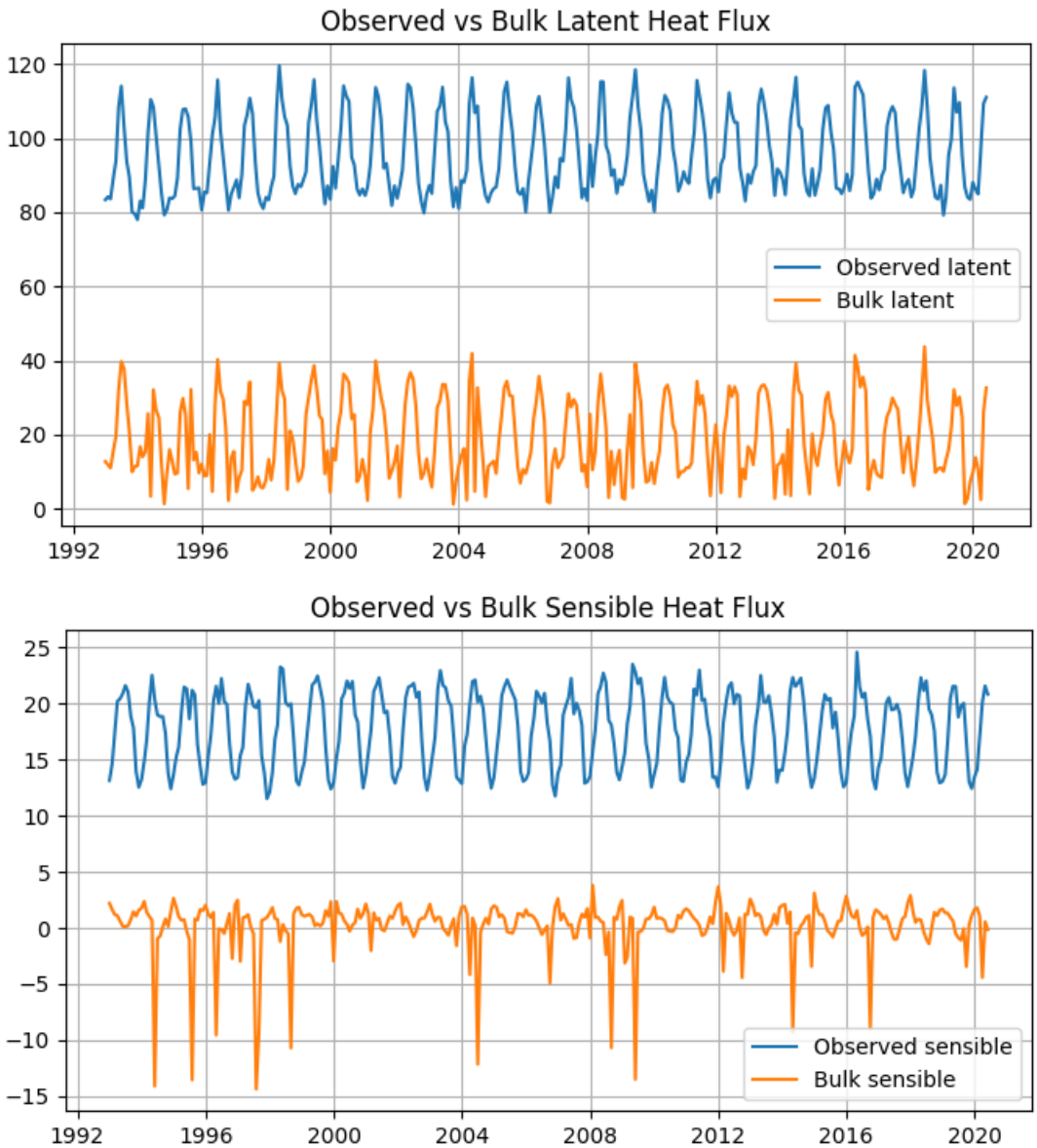

Figure 4: Time-series comparison of bulk and observed latent and sensible heat fluxes.

Bulk formulations capture dominant variability but underestimate nonlinear and density-dependent processes.

6 Linear Regression Driver Importance

6.1 Latent Heat Flux Importance

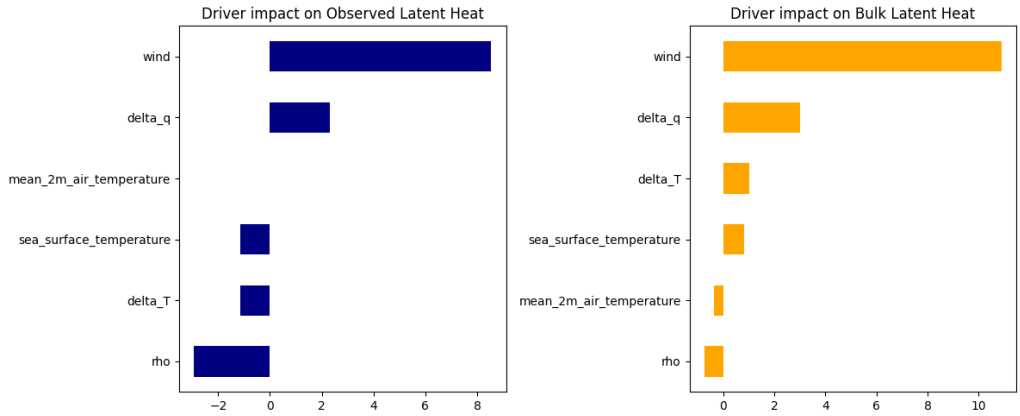

Figure 5: Standardized regression coefficients indicating driver importance for latent heat flux (Observed vs Bulk).

Wind remains dominant, but observed flux shows stronger thermodynamic and density sensitivity.

6.2 Sensible Heat Flux Importance

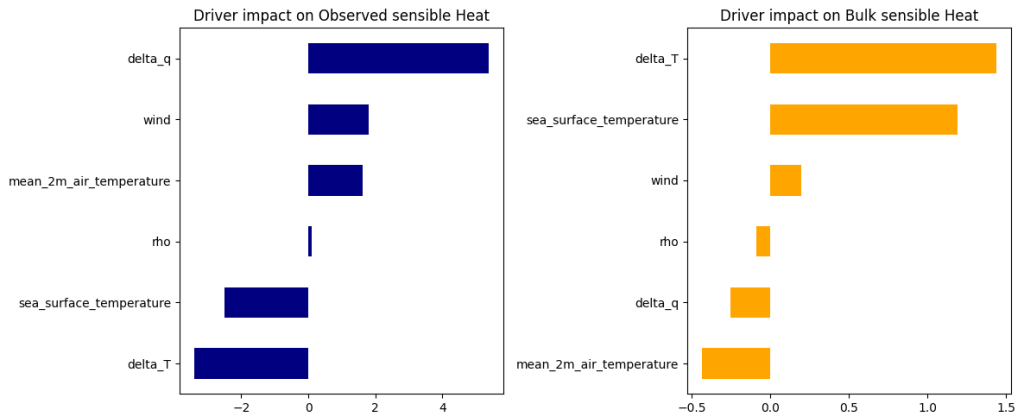

Figure 6: Standardized regression coefficients for sensible heat flux (Observed vs Bulk).

Observed sensible heat shows stronger wind and atmospheric influence than predicted by bulk theory.

7 Response Curve Analysis

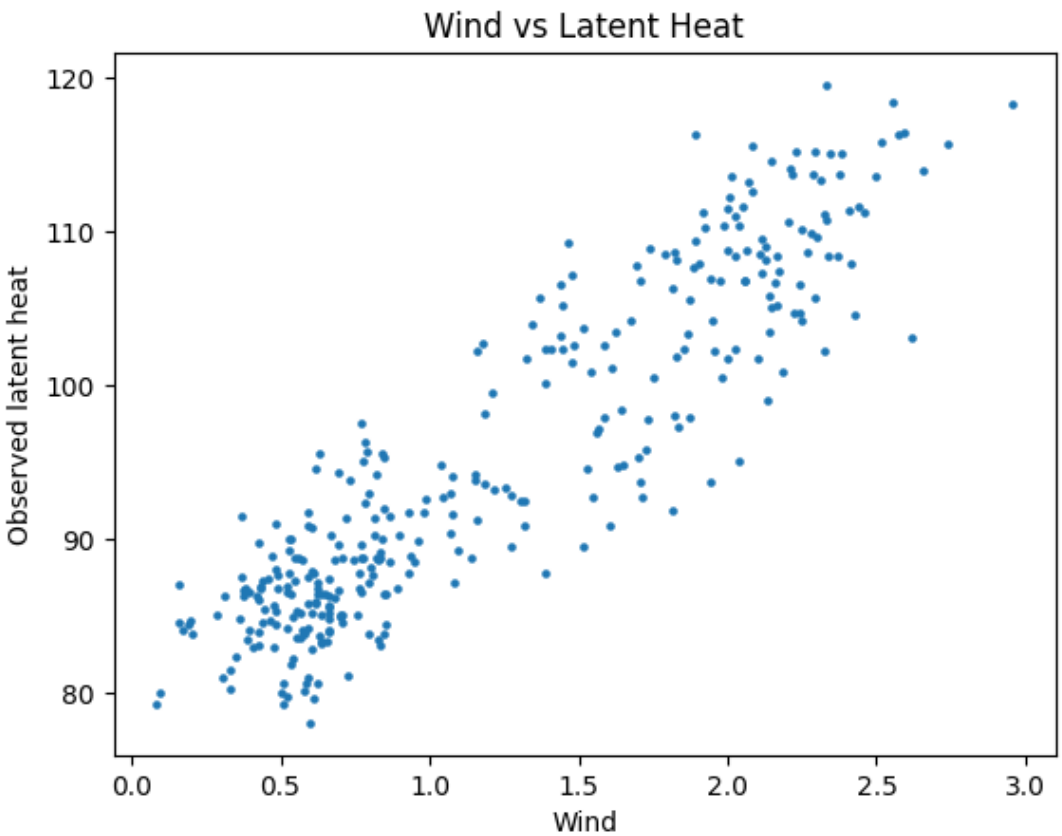

Figure 7: Wind speed vs latent heat flux relationship.

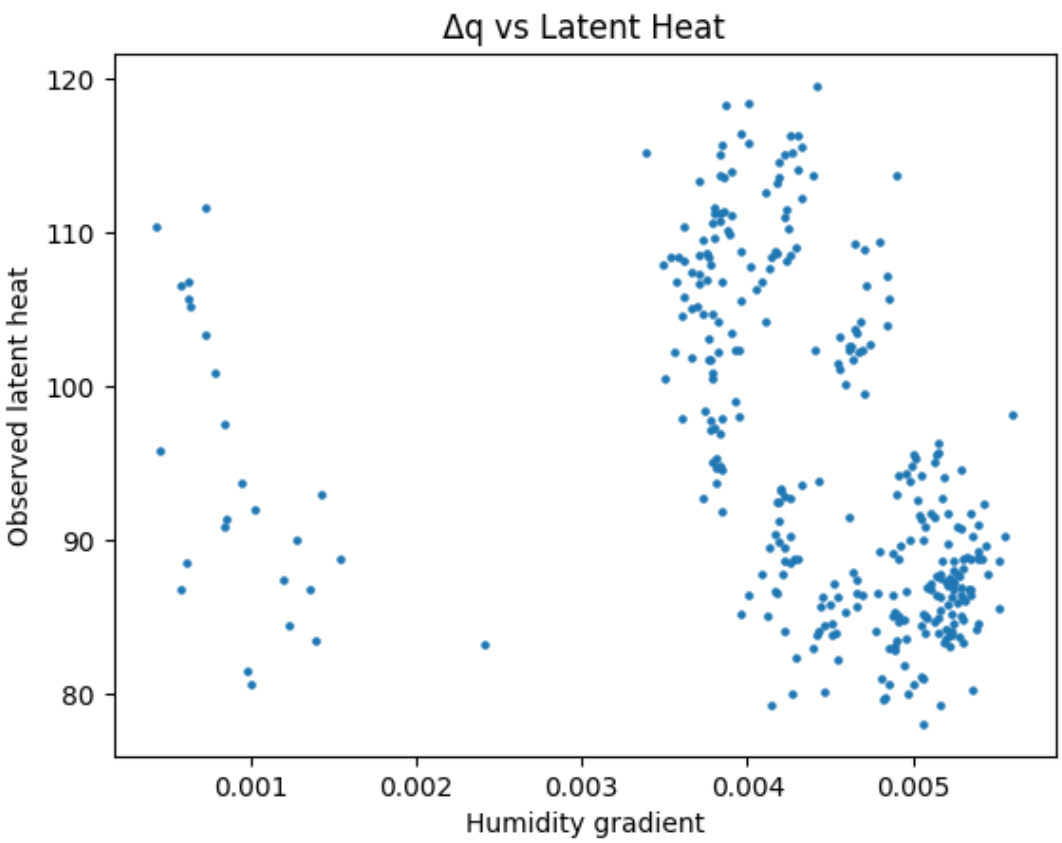

Figure 8: Humidity gradient vs latent heat flux.

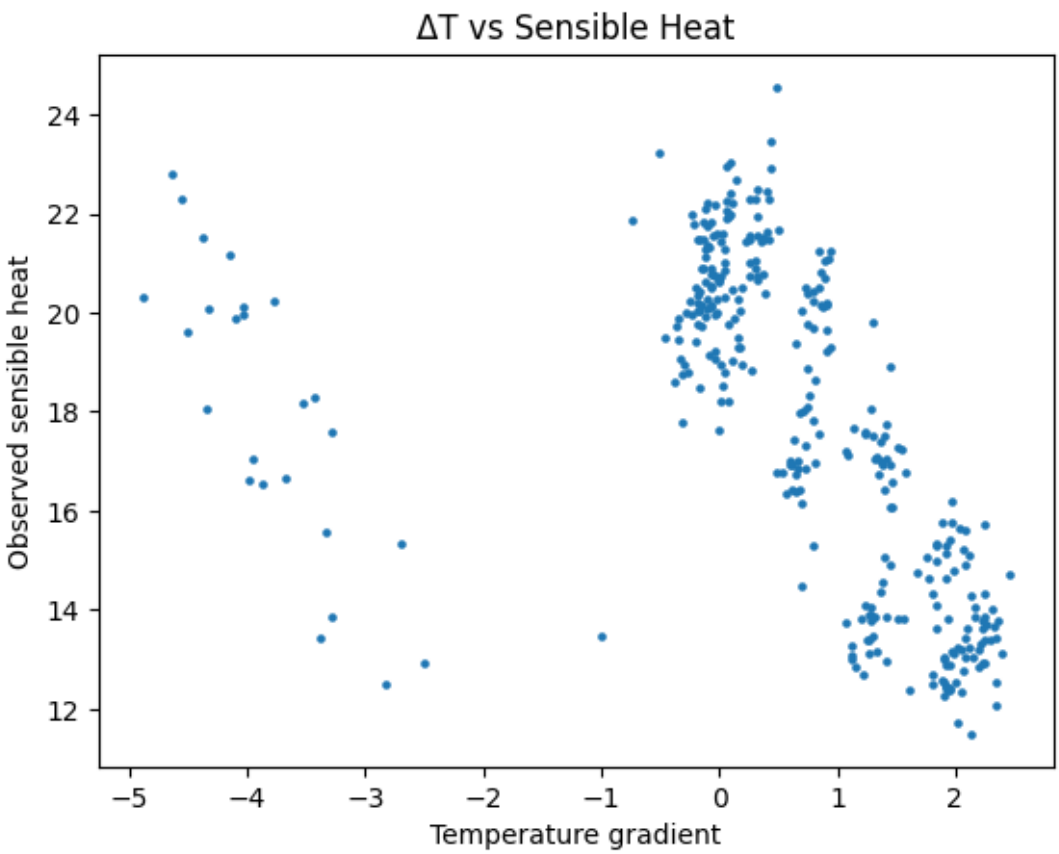

Figure 9: Temperature gradient vs sensible heat flux.

Results confirm expected nonlinear positive relationships.

8 Residual Analysis

8.1 Latent Heat Residuals

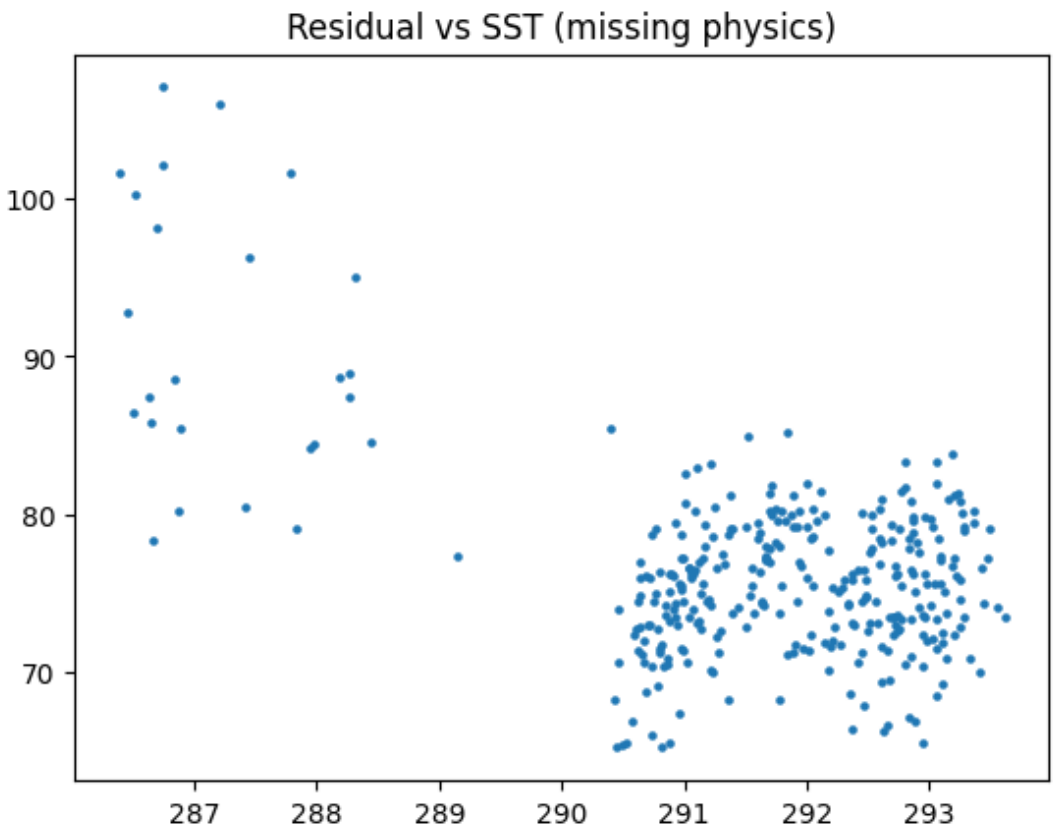

Figure 10: Latent heat residual (Observed and Bulk) vs SST.

Bulk formulas underestimate flux under warm conditions.

8.2 Correlation Heatmap

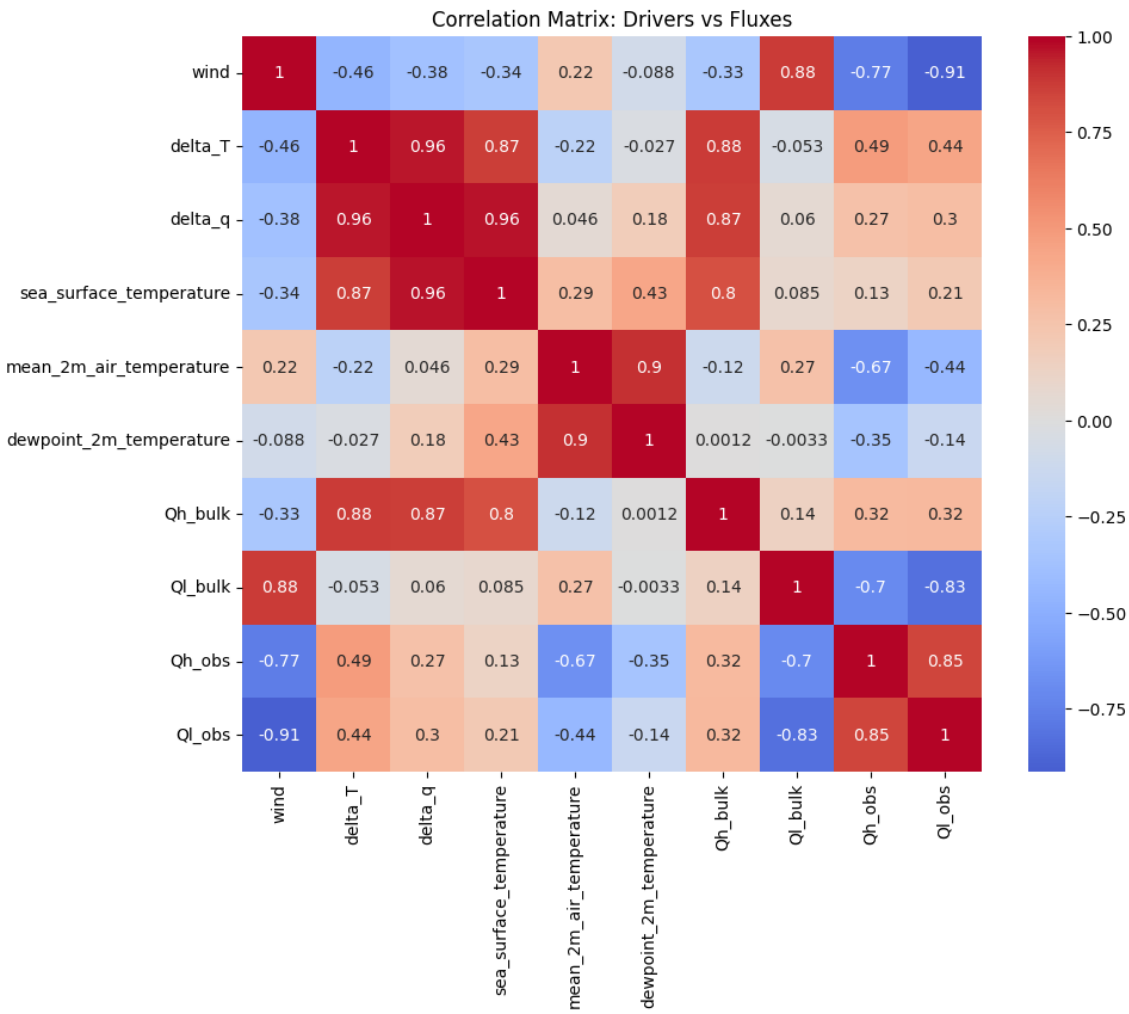

Figure 11: Correlation heatmap: All drivers, gradients, bulk fluxes, observed fluxes.

The heatmap highlights multi-variable dependencies and identifies atmospheric density and pressure as key contributors.

9 Synthesis and Implications for PINN

The EDA phase establishes:

1. Ocean warming drives increasing humidity gradients.
2. Wind dominates turbulent transport.
3. Bulk physics captures primary flux controls.
4. Significant density- and stability-related residuals remain.
5. Residuals exhibit systematic nonlinear structure.

This motivates a Physics-Informed Neural Network to:

- Preserve bulk physical laws.
- Learn boundary-layer corrections.
- Model nonlinear thermodynamic interactions.

Conclusion: The exploratory phase is complete and scientifically validates the transition toward PINN-based modelling.